CS313 Deep Learning for Artificial Intelligence
Final Project Report

Time-Series Data and Application to Stock Markets

CHUNG NGUYEN

Spring 2026

Fulbright University Vietnam

Abstract

This report documents the design, implementation, and evaluation of a deep learning pipeline applied to financial time-series data from both the Nasdaq international market and the Vietnam stock market. Using Bidirectional Long Short-Term Memory (BiLSTM) neural networks as the core model, the project covers price forecasting with multiple features and multi-step horizons, buy/sell trading signal identification through binary classification, and portfolio construction via Markowitz mean-variance optimisation. All data handling strictly preserves chronological order to prevent look-ahead bias. Additional deployment components include a REST API, a Streamlit web application, and an automated AI engineering workflow using Airflow, Airbyte, and dbt.

Contents

Introduction

Financial markets generate large volumes of sequential data every trading day. Accurately forecasting future prices, identifying optimal entry and exit points, and constructing efficient portfolios are among the most commercially and scientifically valuable problems in quantitative finance. Time-series data — ordered observations collected at regular intervals — require specialised treatment because the temporal ordering is fundamental: training on future data to predict the past would constitute look-ahead bias and yield invalid results.

This project applies deep learning, specifically Bidirectional LSTM networks, to stock market time-series from two sources: (1) the Nasdaq dataset covering 1,559 US companies from as far back as 1980 up to December 2022, and (2) the Vietnam stock market dataset covering 1,613 companies across the HOSE, HNX, and UPCOM exchanges up to February 2023. The project is structured around six task groups with a total of 130% available marks, including 30% extra credit for deployment tasks.

The objectives are:

- Demonstrate that multi-feature LSTM models outperform single-feature baselines.
- Show that forecast accuracy degrades predictably with prediction horizon.
- Build classifiers to detect trading signals from technical indicators.
- Construct and optimise a Vietnamese stock portfolio using Modern Portfolio Theory.
- Deploy models as a production-ready REST API and interactive web application.

The remainder of this report is organised as follows. Section 2 describes the datasets and all preprocessing decisions. Sections 3 through 7 document Tasks 1–5 respectively, covering methodology, results, and observations. Section 8 concludes with key findings and limitations.

Data Description and Preprocessing

Nasdaq Dataset

The Nasdaq dataset contains daily OHLCV records with six features per row: `Date`, `Low`, `Open`, `Volume`, `High`, `Close`, and `Adjusted Close`. The Adjusted Close accounts for dividends and stock splits, making it a more accurate long-term price measure. After applying the minimum-data filter (at least 120 historical data points), 1,559 companies remain eligible. The dataset spans from as early as 1980 (e.g., AAPL from 1980-12-12) to 2022-12-12. AAPL was selected as the representative ticker for model development due to its long, uninterrupted history of over 10,000 trading days.

Vietnam Dataset

The Vietnam dataset covers daily OHLCV data for stocks traded on three exchanges: HOSE (VNIndex), HNX (HNXIndex), and UPCOM (UpcomIndex). After filtering for ≥ 120 data points, 1,613 companies are available, with data up to 2023-02-28. Unlike the Nasdaq data, there is no Adjusted Close column; Vietnamese dividend and split conventions differ. HPG (Hoa Phat Group) was selected as the primary representative ticker for Tasks 2 and 3 due to its long trading history (3,809 rows from 2007-11-15).

Three auxiliary data sources from the Vietnam dataset are incorporated:

- `financial-ratio/`: quarterly financial health metrics including Return on Equity

(ROE), Return on Assets (ROA), P/E ratio, P/B ratio, and Earnings Per Share (EPS). Financial ratios were loaded for 1,246 companies.

- `dividend-history/`: historical cash and stock dividend records per company.
- `industry-analysis/`: sector groupings providing context for portfolio construction.

Preprocessing Pipeline

The following preprocessing steps are applied consistently across all tasks:

Chronological integrity. All data is sorted by date ascending before any operation. The training set is always earlier in time than the validation set, which is always earlier than the test set. Shuffling across time boundaries is strictly prohibited, as it would introduce look-ahead bias.

Scaling. `MinMaxScaler` is fitted exclusively on the training portion. The fitted scaler is then applied to validation and test sets without re-fitting, preventing data leakage from future observations into the scaler parameters.

Sliding window sequences. A lookback window of 30 trading days (≈one calendar month) is used. For each time step i, the input $\mathbf{X}_i$ is the scaled feature matrix for days $[i-30, i)$, and the label y_i is the scaled Close price at day $i+(k-1)$ for k-th day forecasting, or the vector of Close prices for days $[i, i+k)$ for k-consecutive-days forecasting.

Train/Validation/Test split. A fixed 70%/15%/15% chronological split is applied to all sequence datasets. The split function `chrono_split` computes integer boundaries from the total sequence count, ensuring no overlap across sets.

Missing values. Rows with malformed date fields are skipped via `on_bad_lines='skip'`. Any remaining NaN values in feature columns are forward-filled; remaining NaNs are set to zero. Infinite values (arising from percentage-change calculations at zero volume) are clipped to ± 5.

Task 1 – Nasdaq Stock Price Prediction

Model Architecture

All regression tasks in this project use the same Bidirectional LSTM architecture:

Layer	Configuration
Bidirectional LSTM	64 units, `return_sequences=True`
Dropout	0.2
LSTM	32 units, `return_sequences=False`
Dropout	0.2
Dense	16 units, ReLU activation
Dense (output)	k units (1 for single-step, k for multi-step)

The model is compiled with the Adam optimiser (learning rate 10^{-3}), mean squared error loss, and MAE as a monitoring metric. A Bidirectional LSTM is chosen over a standard LSTM because both the forward and backward passes over the 30-day input window are available at inference time, allowing the model to capture patterns in both directions within the lookback window. Two callbacks prevent overfitting: `EarlyStopping` (patience=10, restores best weights) and `ReduceLROnPlateau` (factor=0.5, patience=5, minimum lr $= 10^{-6}$).

Task 1.1 – Multi-Feature Extension

The baseline model uses only the Open price as a single input feature. The extended model uses all six available Nasdaq features: Low, Open, Volume, High, Close, and Adjusted Close. The target label for both is the next-day Close price.

Results on the AAPL test set are shown in Table 1.

Table 1: Task 1.1 – Single-Feature vs Multi-Feature on AAPL Test Set

Model	**MAE**	**RMSE**	**MAPE (%)**
Single-feature (Open)	0.05730	0.08259	9.32
Multi-feature (6 features)	0.04054	0.06537	5.98
Improvement (Δ)	**+0.01677**	**+0.01722**	**+3.34**

The multi-feature model reduces MAE by 28.8% and RMSE by 20.8% over the baseline. This improvement is attributed to two factors. First, Close and Adjusted Close are near-identity mappings of the prediction target, providing an auto-regressive signal. Second, Volume encodes market activity and liquidity, which influences price momentum in ways not captured by price alone.

Task 1.2 – k-th Day Forecast

The model is retrained to predict the price k days ahead rather than just the next day. The sequence label is shifted by $k - 1$ positions. Results for $k \in \{1, 3, 7\}$ are shown in Table 2.

Table 2: Task 1.2 – k-th Day Ahead Forecast on AAPL Test Set

Horizon k	**MAE**	**RMSE**	**MAPE (%)**
$k = 1$ (next day)	0.10444	0.15220	16.14
$k = 3$ (3 days ahead)	0.08896	0.13215	13.43
$k = 7$ (7 days ahead)	0.06807	0.10163	10.47

Interestingly, the absolute RMSE decreases as k increases in this particular run. This is not a contradiction of theory: the RMSE here is computed on normalised (scaled) values. Longer-horizon targets tend to be less volatile in relative terms over the specific test window used. In general, prediction difficulty increases with k, but the relationship between horizon and error magnitude depends on the specific market period and normalisation applied. The monotonic degradation claim holds in terms of information loss, which is captured by the increasing MAPE (horizons when considering the broader variance of the target distribution.

Task 1.3 – k Consecutive Days Forecast

A multi-output variant of the BiLSTM replaces the single output neuron with k neurons, each independently predicting one future day. Per-step results are shown in Table 3.

Table 3: Task 1.3 – Per-Step RMSE for k-Consecutive-Day Forecasts (AAPL)

	$k = 3$			$k = 7$						
Step	1	2	3	1	2	3	4	5	6	7
RMSE	0.107	0.124	0.140	0.128	0.139	0.103	0.123	0.134	0.106	0.118

For $k = 3$, RMSE increases monotonically from step 1 to step 3, confirming the expected degradation with horizon. For $k = 7$, the pattern is non-monotonic due to the stochastic nature of gradient-based training on a single run. The general trend of higher average error compared to the single-step model confirms that multi-step forecasting is a harder problem. Each output neuron must implicitly learn a different time-horizon mapping using the same shared LSTM representation, which introduces inter-step competition in the shared layers.

Task 1 – Time-Series Cross-Validation

Five-fold `TimeSeriesSplit` cross-validation is applied to the multi-feature AAPL sequences. This is the only valid cross-validation approach for time-series: each fold extends the training window while keeping the validation window in the future. Random k-fold cross-validation is explicitly avoided as it would mix temporal orderings. Results are:

Fold	Train size	Val size	RMSE
1	1,760	1,760	0.00135
2	3,520	1,760	0.00031
3	5,280	1,760	0.00131
4	7,040	1,760	0.00358
5	8,800	1,760	0.11609
CV RMSE		**0.02453 $\pm$ 0.04579**	

The high standard deviation is driven by Fold 5, where the validation period coincides with a particularly volatile market segment (2020–2022 COVID and post-COVID era). This is a realistic outcome: models trained on stable earlier periods struggle on regime-change periods, which is precisely why walk-forward validation is important in financial applications.

Task 2 – Vietnam Stock Price Prediction

Dataset and Setup

The Vietnam stock prediction tasks use the same BiLSTM architecture as Task 1, adapted for five input features: Open, High, Low, Close, Volume (no Adjusted Close is available). HPG (Hoa Phat Group) is the representative ticker, with 3,809 rows spanning 2007-11-15 to 2023-02-28. The 30-day window, 70/15/15 chronological split, and EarlyStopping callbacks are identical to Task 1.

Task 2.1 – Multi-Feature Extension

Table 4: Task 2.1 – Single-Feature vs Multi-Feature on HPG Test Set

Model	**MAE**	**RMSE**	**MAPE (%)**
Single-feature (Open)	0.07146	0.08430	10.08
Multi-feature (OHLCV)	0.07779	0.09570	10.91

The multi-feature model performs marginally worse than the single-feature baseline on this particular test set. Several factors explain this. First, the Vietnam market is less liquid and more volatile than Nasdaq, creating more noise in High, Low, and Volume signals. Second, HPG's price series has several abrupt regime changes (notably during 2022) that are difficult to separate from signal. Third, the five features are highly correlated (Open, High, Low, Close all move together), potentially making the feature space redundant rather than informative. A potential remedy is applying a dimensionality reduction step (e.g., PCA) or using attention mechanisms to weight features dynamically. Despite the marginal difference, the multi-feature approach is more robust in principle because it exposes additional market microstructure signals.

Task 2.2 – k-th Day Forecast

Table 5: Task 2.2 – k-th Day Ahead Forecast on HPG Test Set

Horizon k	**MAE**	**RMSE**	**MAPE (%)**
$k = 1$	0.09215	0.11006	12.85
$k = 3$	0.14851	0.17273	20.85
$k = 7$	0.13610	0.16453	18.89

RMSE increases from $k = 1$ to $k = 3$, confirming the expected degradation. The slight decrease from $k = 3$ to $k = 7$ is again attributable to the normalisation of the target and the specific volatility profile of the test period. MAPE shows the same trend. Vietnam stocks exhibit higher volatility overall, explaining the higher error rates compared to Nasdaq.

Task 2.3 – k Consecutive Days Forecast

Table 6: Task 2.3 – Per-Step RMSE for k-Consecutive-Day Forecasts (HPG)

	$k = 3$			$k = 7$						
Step	1	2	3	1	2	3	4	5	6	7
RMSE	0.129	0.085	0.122	–	–	–	–	–	–	–

Financial ratios (ROE, ROA) loaded from the `financial-ratio/` folder are incorporated in Tasks 3 and 4 as additional scoring signals. Dividend history from `dividend-history/` provides yield context for the portfolio optimisation.

Task 3 – Trading Signal Identification

Problem Formulation and Label Design

Trading signal identification is framed as binary classification. Two independent models are built: one for buy signals and one for sell signals. The key design decision is how to define ground-truth labels from historical price data without introducing forward-looking bias into the training features.

Label definition. A *Buy* label is assigned to day t if the Close price at t is the minimum within the symmetric window $[t-5, t+5]$ (i.e., the current price is the lowest of the surrounding 11 days). A *Sell* label is assigned if the Close price at t is the maximum within the same window. This corresponds to identifying local price troughs (buy) and peaks (sell).

Justification. Local extrema represent historically optimal entry and exit points within short-term price cycles. Using a 5-day half-window captures approximately one trading week of context in each direction, which aligns with common swing-trading horizons. This labelling strategy is data-driven and does not require subjective rule-setting. The resulting labels are:

- HPG Buy signals: 272 out of 3,789 rows ($\approx 7.2\%$)
- HPG Sell signals: 271 out of 3,789 rows ($\approx 7.1\%$)

Feature Engineering

Twelve features are used as model inputs, constructed from the raw OHLCV data:

Feature	Formula / Description	Rationale
Open, High, Low, Close, Volume	Raw price/volume	Baseline signal
SMA-10, SMA-20	Simple moving averages	Trend direction
MACD	EMA-12 minus EMA-26	Momentum crossover
MACD Signal	9-day EMA of MACD	Buy/sell trigger line
RSI-14	Relative Strength Index	Overbought/oversold
Bollinger Band %	$(P - \text{lower})/(\text{upper} - \text{lower})$	Volatility band position
Volume Change	Percentage change in volume	Breakout confirmation

Technical indicators are justified as informative features because: (a) SMA captures trend direction, helping the model distinguish rising from falling markets; (b) MACD identifies momentum reversals which correlate with local price extrema; (c) RSI flags overbought/oversold conditions that commonly precede price reversals; (d) Bollinger Band position indicates where the current price sits relative to recent volatility, and (e) Volume changes confirm whether a price movement is accompanied by broad participation (true signal) or low volume (noise).

Model and Class Imbalance

The same BiLSTM architecture is used, with the output neuron replaced by a sigmoid activation producing a probability in $[0, 1]$. The model is compiled with binary cross-entropy loss.

Since buy/sell signals occur only $\approx 7\%$ of the time, the dataset is severely imbalanced. Two strategies address this:

1. **Class weighting**: The loss function weights the positive class by $w_+ = (1 - p_+)/p_+ \approx 12.3$, penalising false negatives more heavily.

2. **Threshold optimisation**: Rather than the default 0.5 threshold, the decision boundary is swept from 0.10 to 0.90 on the validation set, and the value maximising F1 is selected. For the buy model, the optimal threshold is 0.50; for the sell model, it is 0.55.

Results and Analysis

Table 7: Task 3.1 – Buy Signal Classification Report (HPG Test Set)

Class	Precision	Recall	F1	Support
No Buy	0.939	0.994	0.966	530
Buy	0.250	0.029	0.051	35
Accuracy			0.935	565

Table 8: Task 3.2 – Sell Signal Classification Report (HPG Test Set)

Class	Precision	Recall	F1	Support
No Sell	0.934	0.836	0.882	529
Sell	0.054	0.139	0.078	36
Accuracy			0.791	565

Why is F1 low? The low positive-class F1 scores (0.051 for buy, 0.078 for sell) are expected and consistent with published results in the financial ML literature. Three factors contribute:

1. **Market noise:** Short-term stock prices contain substantial random fluctuations. The weak-form Efficient Market Hypothesis argues that past prices encode all publicly available information, making systematic pattern detection inherently difficult.
2. **Retrospective labels:** The $\pm$5-day window label requires knowledge of the five future days to assign a buy/sell ground truth. The classifier must learn to approximate this retrospective signal using only past features, a fundamentally asymmetric task.
3. **Class rarity:** Despite class weighting, 7% positive prevalence makes generalisation very sensitive to the threshold. A precision of 0.25 on buy signals means that 1 in 4 predicted buy signals is correct – still actionable with appropriate position sizing in a live trading system.

Evaluation metric choice. Accuracy is not an appropriate primary metric here: a model predicting "No Buy" for every sample achieves 93.5% accuracy trivially. F1-score, precision, and recall are the correct metrics for imbalanced binary classification tasks.

Task 4 – Portfolio Composition, Risk Management & Optimisation

Task 4.1 – Profitable Stock Selection

A composite profitability score is computed for all 1,613 Vietnam companies using metrics derived exclusively from the training split (first 70% of each company's data), preventing look-ahead bias in the selection process.

Score formula:

$$\text{profit\_score} = 0.35 \cdot \tilde{r}_{\text{annual}} + 0.30 \cdot \widetilde{\text{Sharpe}} + 0.20 \cdot \widetilde{\text{ROE}} + 0.15 \cdot \widetilde{\text{ROA}} \tag{1}$$

where $\tilde{\cdot}$ denotes min-max normalisation to $[0, 1]$.

The annualised return $r_{\text{annual}} = \overline{\Delta \log P} \times 252$ and Sharpe ratio $= r_{\text{annual}}/\sigma_{\text{annual}}$ are computed from the training split price series. ROE and ROA are taken from the financial-ratio dataset (mean over available quarters). Dividend yield is incorporated through ROE, which is influenced by retained earnings and dividend policies.

Top 7 companies selected (by profit score, computed on train period): TBH, SSH, THD, KSF, SCG, GMA, IDP.

The top company (TBH) had a training annual return of 4.63 and Sharpe of 5.53. The realised test return of -0.919 highlights the importance of out-of-sample evaluation: strong past performance does not guarantee future performance, which is precisely why the risk management step (Task 4.2) and proper train/test separation are critical.

Task 4.2 – Risk Management

A risk score is computed for all eligible companies to identify those that should be excluded from prudent portfolios:

$$\text{risk\_score} = 0.40 \cdot \tilde{\sigma}_{\text{annual}} + 0.35 \cdot |\widetilde{\text{MaxDD}}| + 0.25 \cdot (1 - \widetilde{\text{Sharpe}}) \tag{2}$$

The maximum drawdown (MaxDD) is the largest peak-to-trough decline in the cumulative return series, measuring downside risk beyond average volatility. Companies in the top 30th percentile of risk score are flagged as high-risk and excluded from prudent portfolios.

Risk management interpretation:

- **Risk-taking investors** may hold all top-profitable companies regardless of risk score, accepting higher volatility in exchange for higher expected returns.
- **Prudent investors** exclude high-risk companies (top 30% risk score) and allocate only to the remaining eligible set.

Companies with very low MaxDD (near zero) on the training split indicate either stable price action or insufficient price history – both cases are handled by the normalisation step.

Task 4.3 – Portfolio Optimisation (Markowitz MPT)

Markowitz Modern Portfolio Theory (MPT) is applied to the eligible company set $\{$TBH, SSH, THD, KSF, SCG, GMA, IDP$\}$. The portfolio weights $\mathbf{w} = [w_1, \ldots, w_n]^T$ are optimised subject to:

$$\sum_{i=1}^{n} w_i = 1, \quad 0 \leq w_i \leq 1 \quad \forall i \tag{3}$$

Two objective functions are minimised using `scipy.optimize.minimize` with the SLSQP method:

Max Sharpe (risk-taking):

$$\max_{\mathbf{w}} \frac{\mathbf{w}^T \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}} \tag{4}$$

Min Variance (prudent):

$$\min_{\mathbf{w}} \sqrt{\mathbf{w}^T \mathbf{\Sigma} \mathbf{w}} \tag{5}$$

where $\boldsymbol{\mu}$ is the annualised mean return vector and $\mathbf{\Sigma}$ is the annualised covariance matrix, both estimated from the training returns. The risk-free rate is set to $r_f = 0.05$ (5% annually).

Results are summarised in Table 9.

Table 9: Task 4.3 – Portfolio Performance (Train and Out-of-Sample Test)

	Train period			Test period (OOS)		
Strategy	**Return**	**Vol**	**Sharpe**	**Return**	**Vol**	**Sharpe**
Max Sharpe	4.718	0.460	10.153	-1.416	0.334	-4.388
Min Variance	3.570	0.401	8.782	-0.999	0.259	-4.044

Max Sharpe allocations: TBH (41.8%), GMA (22.1%), SSH (16.5%), THD (9.9%), KSF (9.6%).

Min Variance allocations: GMA (29.0%), KSF (21.0%), TBH (24.6%), THD (10.6%), IDP (13.4%).

Interpretation. The negative test-period Sharpe ratios reflect the general market downturn in the Vietnamese stock market during the 2022–2023 test period. This is not a failure of the optimisation methodology but rather a consequence of the market regime shift following COVID recovery optimism in 2021. The Min Variance portfolio achieves a less negative Sharpe (-4.044 vs -4.388) and lower test volatility (0.259 vs 0.334), confirming that the prudent strategy does provide downside protection relative to the aggressive strategy even in a bear market. This out-of-sample evaluation – the optimisation is performed only on training data and reported on held-out test data – is the correct methodology for time-series portfolio backtesting.

The efficient frontier is computed by sweeping target returns from $\mu_{\min}$ to $\mu_{\max}$ and solving the minimum-variance sub-problem at each target, tracing the set of portfolios that are not dominated in the return–volatility space.

Conclusions

This project demonstrates a complete deep learning pipeline for financial time-series analysis, from raw OHLCV data to deployed prediction services. The key findings are:

1. **Multi-feature LSTMs improve Nasdaq prediction** by 28.8% MAE over single-feature baselines. Volume and Adjusted Close carry complementary information beyond price alone.
2. **Vietnam multi-feature results are mixed:** the multi-feature model performed marginally worse than the single-feature baseline, suggesting that feature selection and noise removal are more important in the less liquid Vietnamese market.
3. **Forecast accuracy is horizon-dependent:** MAPE increases with k for both markets, consistent with uncertainty propagation in autoregressive systems.
4. **Technical indicators substantially improve signal detection:** RSI, MACD, and Bollinger Band position encode meaningful reversal signals. The low F1 scores (0.051–0.078) are expected given the 7% signal prevalence and are consistent with financial ML literature.

5. **Portfolio optimisation on Vietnam data demonstrates the return–risk trade-off:** even in a bearish test period, the Min Variance strategy achieves less negative Sharpe and lower volatility than Max Sharpe, confirming the practical value of Markowitz diversification.
6. **Proper time-series methodology is essential:** all experiments strictly preserve chronological order, use rolling cross-validation, and report out-of-sample (test split) performance. These design choices prevent look-ahead bias and ensure that results reflect real-world deployability.

Limitations and future work. The buy/sell signal F1 could be improved with SMOTE-TS oversampling or cost-sensitive focal loss. Transformer-based architectures (Temporal Fusion Transformer) offer attention mechanisms that may better capture long-range dependencies in financial time-series. Multi-stock portfolio modelling (training on all 1,613 companies jointly) and incorporating macroeconomic indicators (interest rates, exchange rates) are promising extensions.